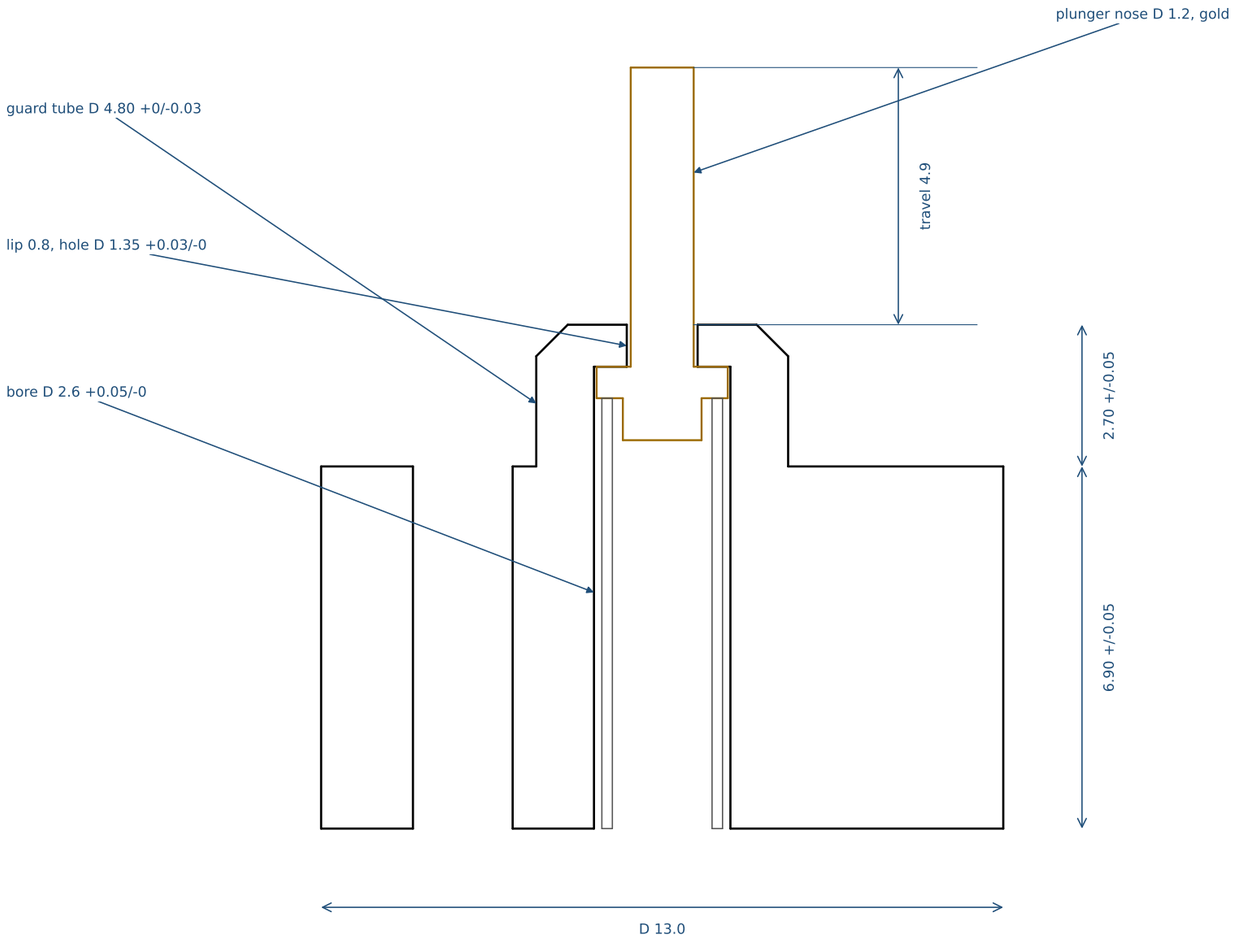
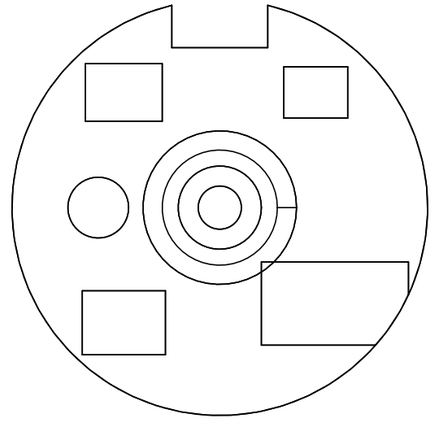
SECTION y=0 (backplate back face at z=0); plunger shown free (cap off), spring grey




TOP VIEW (+Y = 12 o'clock, toward the splice cavity)



**NOTES:**

1. Sits in the RS-HUB-01 rev B centre relief (D 14.0 x 9.1) on top of RS-PCB-12L-HUB; frame z=0 = backplate back face. Body D 13.0 x 6.90 +/-0.05.
2. GUARD TUBE = the spigot: D 4.80 +0/-0.03, top at z 2.1 (2.70 +/-0.05 above the body), 0.6 x 45 lead-in. It pilots in the back of the E-Z Lok thread (minor D 4.98..5.26) and its top is a HARD STOP 0.56 below the lowest position a correct lit-cap stud can reach with every tolerance stacked: no screw of any length can crush the contact.
3. Plunger bore D 2.6 +0.05/-0 from the underside to 0.8 below the guard top; D 1.35 +0.03/-0 through the 0.8 lip. The lip retains the plunger (flange D 2.5 x 0.6) with the cap off; nose tip then stands at z 7.0 (4.9 of travel down to the guard top).
4. SPRING (calc, catalogue SKU to be picked): OD 2.3, wire 0.2, free length 11.2, 13 coils closed ends, about 0.15 N/mm, stainless. Solid height must be <= 3.3. It sits on a flat gold pad on the PCB and on the plunger pilot (D 1.5 x 0.8); it carries the 20 mA.
5. Return pin guide bore D 1.9 thru at (-3.8, 0.0) for Mill-Max 0919-0-15-20-89-14-11-0 (SMT on the PCB; working height 7.5 against the insert's back end). Confirm bore = barrel OD + 0.10 against the Mill-Max drawing.
6. Underside pockets: CCR 4.6 x 2.6 x 1.4, D1 2.6 x 2.0, Z1 2.0 x 1.6, F1 2.4 x 1.8; wire notch 3.0 wide toward +Y. Drop spring + plunger into the bore, bond the carrier to the PCB with 2 dots of epoxy (none in the bore). Test: plunger returns freely from full depth 10 times.

**RECORD SCONCE | HUB CONTACT CARRIER + GUARD TUBE, plunger and spring**

PART NO: RS-HUB-12L-CARRIER REV: B DATE: 2026-11-11 SUPERSEDES: rev A, A.1 (do not build or quote)
MATERIAL: CARRIER: PEEK (preferred) or PA66-GF30, machined; prototype SLA Formlabs High Temp / Tough 2000. Insulator. PLUNGER: brass C360 or BeCu, turned, nose D 1.2 with R 0.3 crown. SPRING: catalogue compression spring, see note 4.
FINISH: CARRIER as machined, no metal inserts. PLUNGER: 1.3 um hard gold over 2.5 um nickel all over.
UNITS: mm [in] THIRD-ANGLE PROJECTION SCALE: as shown
GENERAL TOLERANCES: x.x +/-0.3 x.xx +/-0.1 angles +/-0.5 deg unless noted